

DEVELOPMENT OF THE PALATE

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TASK 1: IDENTIFY, STUDY AND PASTE THE DIAGRAMS

Development of Palate/Palatogenesis

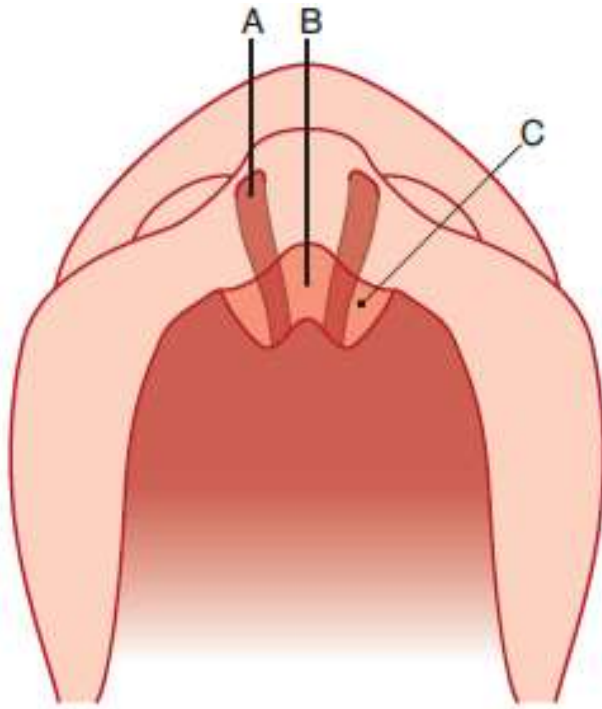


Fig. 18.1 The state of development of the palate by the 6th week of intrauterine life.

A – Primitive nasal cavities: Early right and left nasal sacs.

B – Primary nasal septum: A thin vertical partition formed by the fusion of the medial nasal processes.

C – Primary palate: A small triangular plate of tissue derived from the intermaxillary segment.

Development of Palate/Palatogenesis

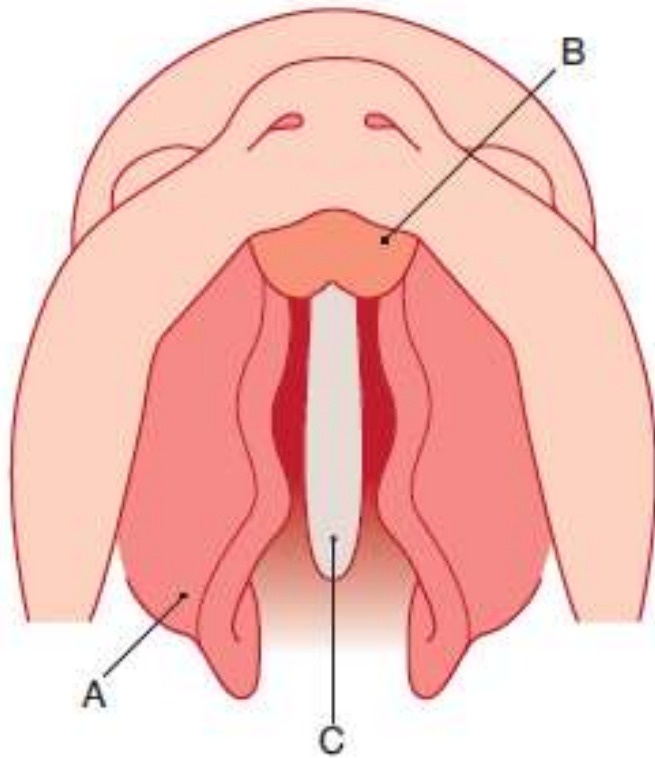


Fig. 18.2 The development of the palate during the 6th week of intrauterine life.

A – Lateral palatal shelves: Outgrowths from the maxillary processes on each side of the tongue. Initially, they grow vertically downward.

B – Primary palate: Formed earlier (fusion of medial nasal processes); lies anterior to the incisive foramen.

C – Secondary nasal septum: A downward mesenchymal extension from the frontonasal prominence, which will later fuse with the secondary palate.

Development of Palate/Palatogenesis

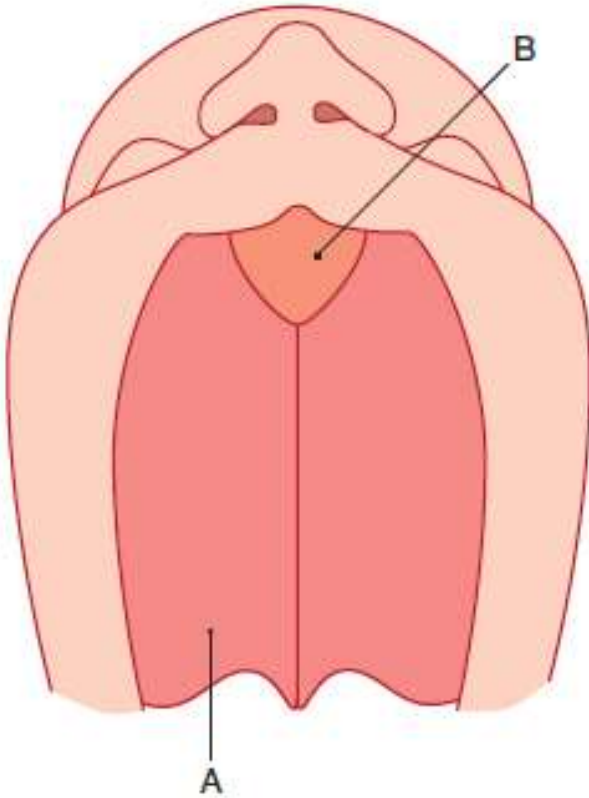


Fig. 18.4 The state of development of the palate during the 8th week of intrauterine life.

A – Palatal shelves: elevate and grow medially and fuse with each other and with the posterior border of the primary palate.

B – Primary palate: lies anterior to the incisive foramen. This fusion creates a Y shaped suture.

EXTRA READING MATERIAL



Frontonasal (single)

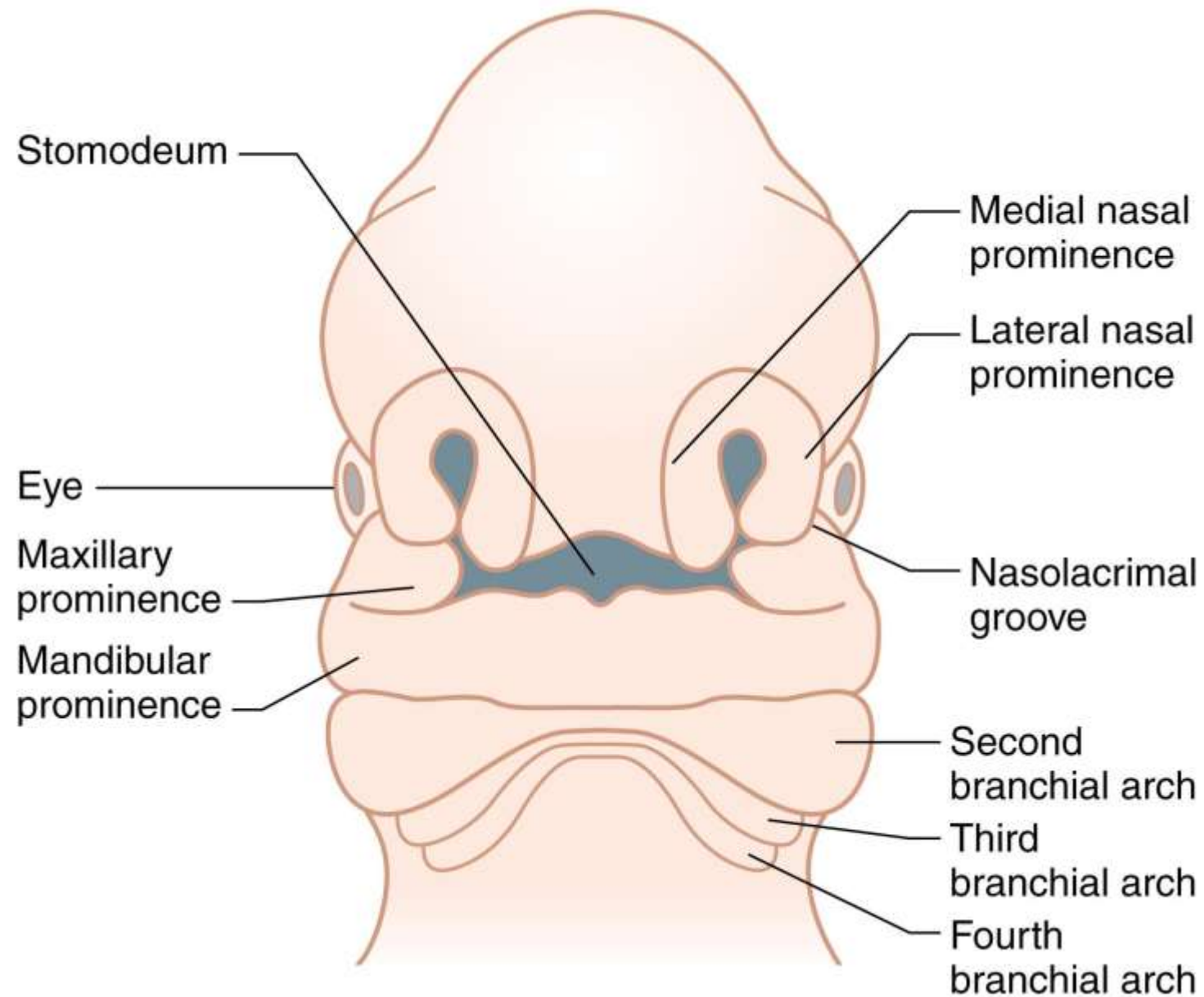
Formed from the mesenchyme in front of dev. brain



Maxillary and mandibular processes (paired). Derived from the 1st pharyngeal arch



Proliferation of mesenchyme from FNP around openings of nasal pit will form medial and lateral nasal processes.



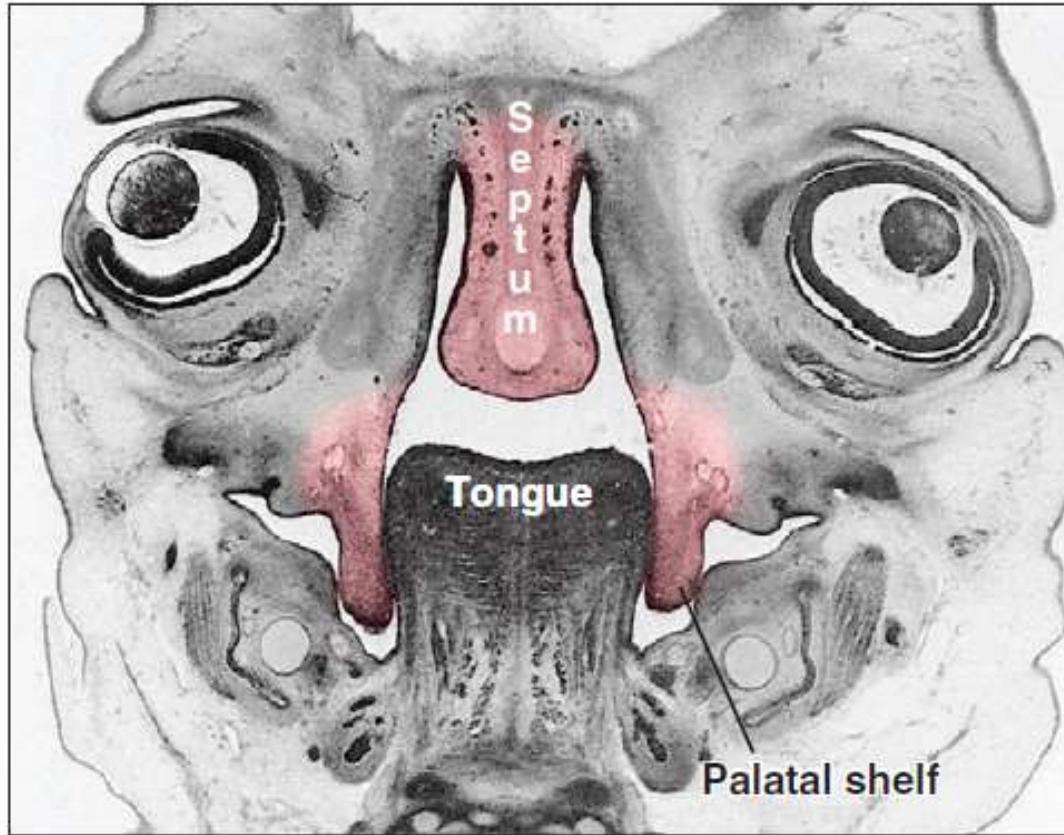
primary palate that is the caudal aspect of one of the processes involved in facial development – the frontonasal process

secondary palate is derived from two medial extensions of the maxillary processes of the developing face – called palatal shelves

Palatogenesis timeline: 6th to 12th week

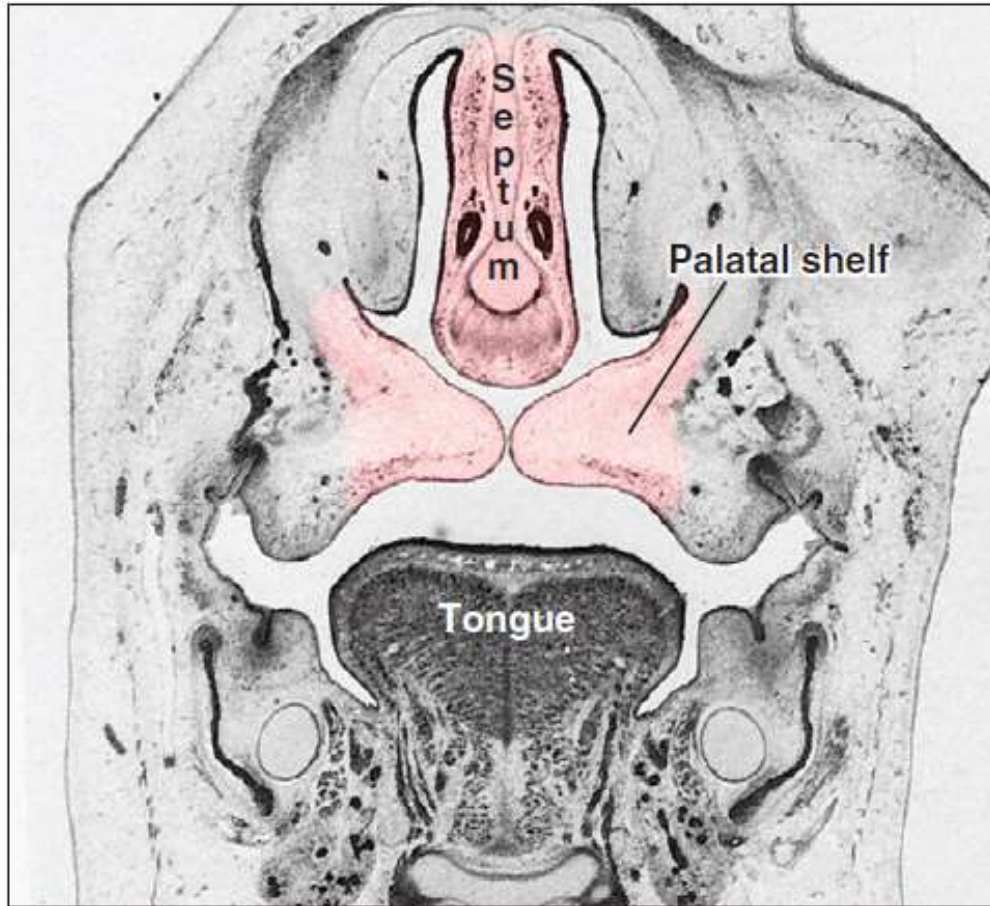
During the 6th week, the two medial nasal processes fuse in the midline, forming the **intermaxillary segment**. The intermaxillary segment extends backward between the maxillary processes to form a small triangular wedge of tissue — this is the **primary palate**.

Definitive palate is made by fusion of secondary nasal septum (from frontonasal process) with the two palatal shelves (extensions of the maxillary process) and the primary palate anteriorly. This fusion is complete by 12th week.



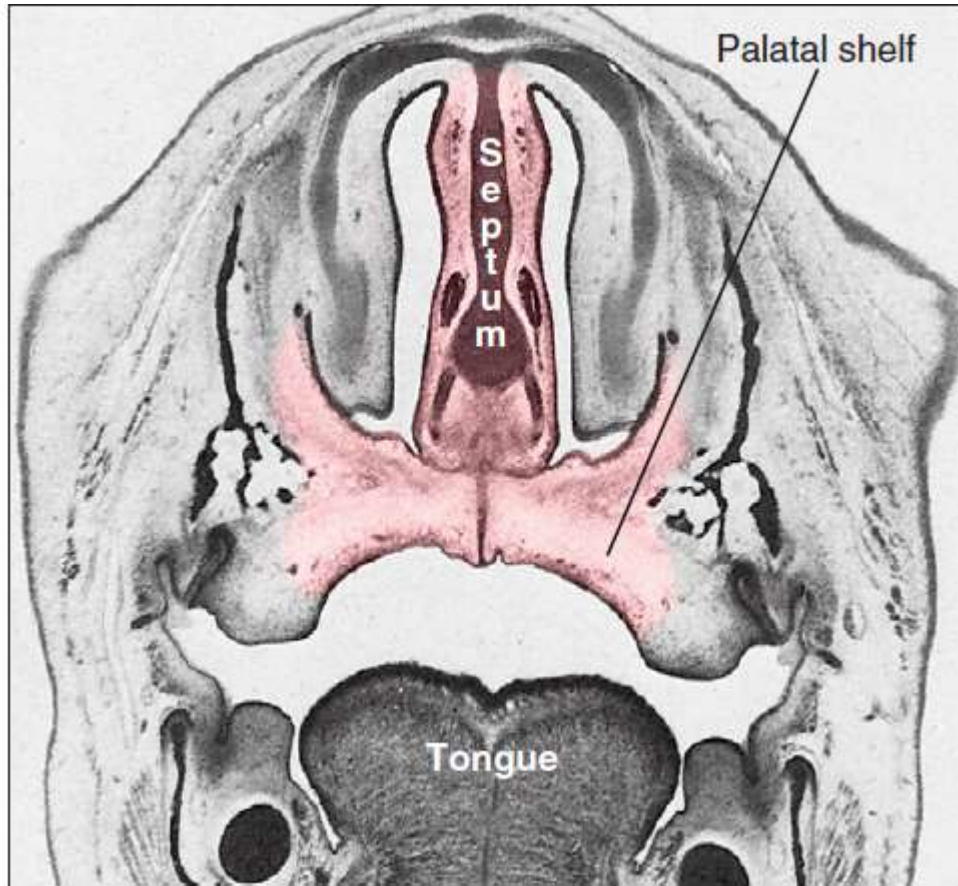
At 7 weeks the palatine shelves grow vertically downward on either side of the developing tongue.

The secondary nasal septum divides the primitive nasal cavities into left and right.



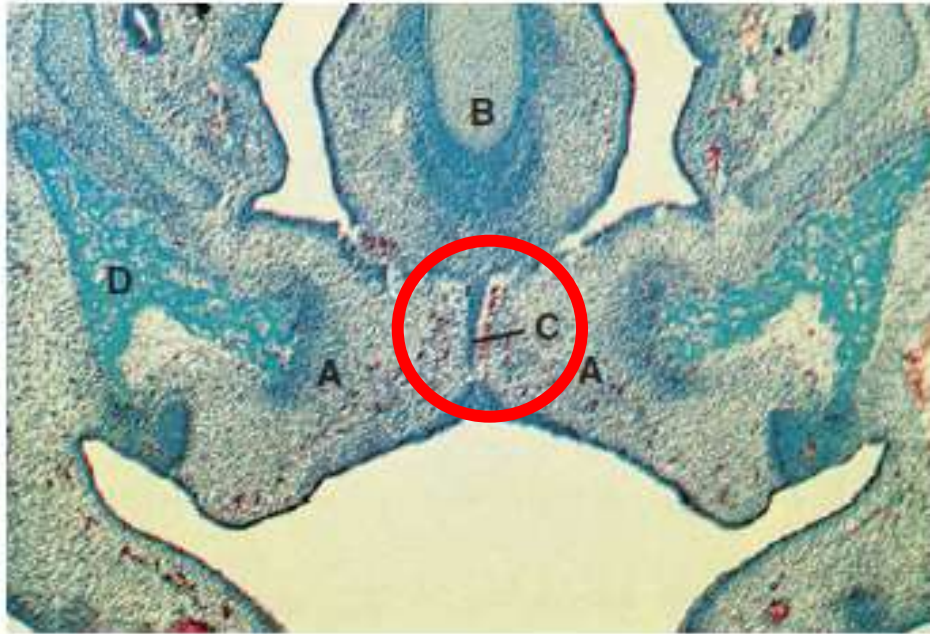
At 8 weeks the palatine shelves orient horizontally and elevate to meet medially.

The secondary nasal septum grows downward to fuse with these palatine processes.



At 9 weeks the fusion is complete of all three structures, resulting in formation of secondary palate and the separation of nasal cavity and oral cavity from each other.

Fusion of shelves



C- Once the palatal shelves have elevated, they contact each other (initially in the middle third of the palate) and adhere by means of a 'sticky' glycoprotein, which coats the surface of the medial edge epithelia of the shelves **midline epithelial seam**.

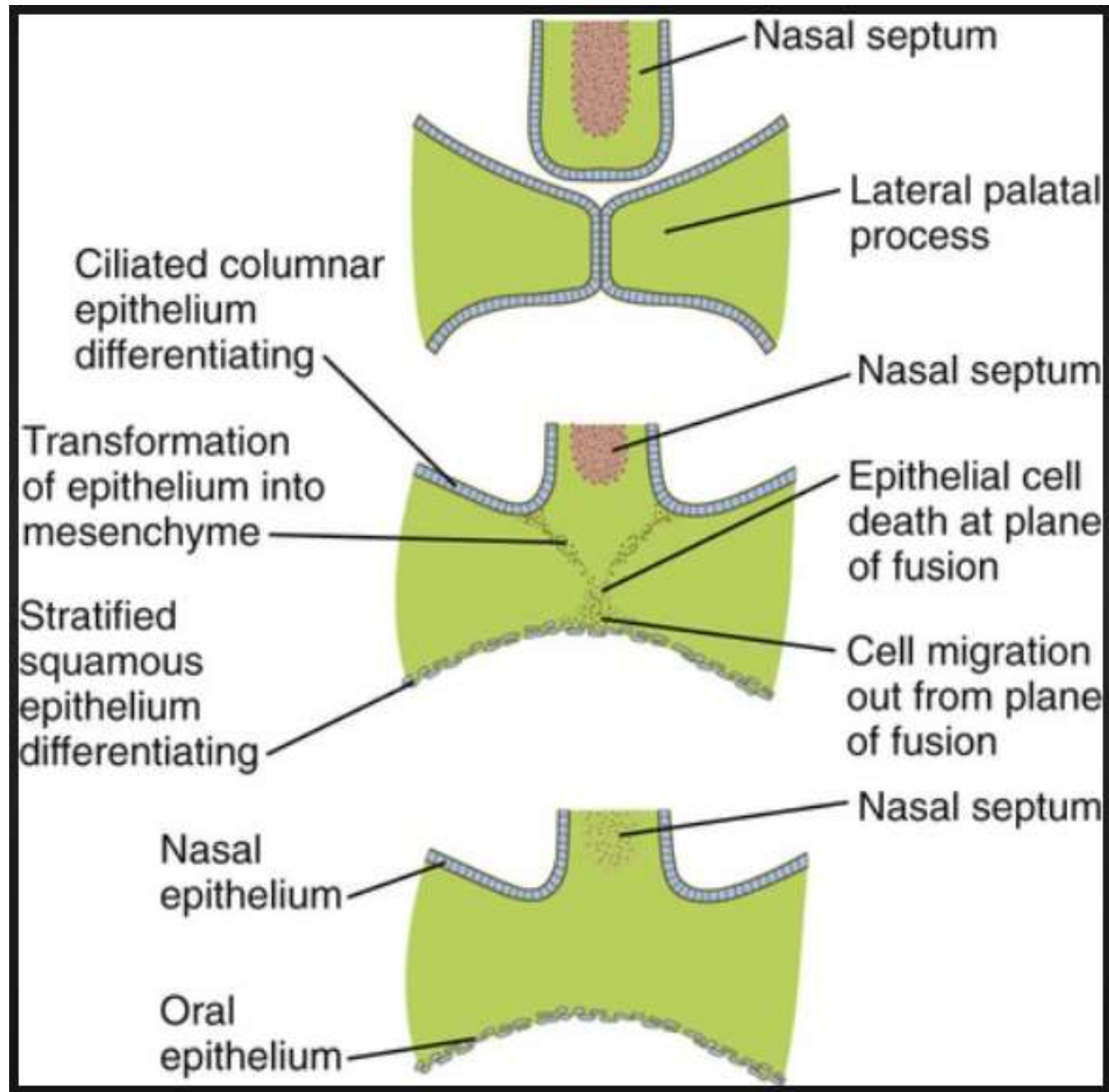
1. At a later stage of fusion, with **migration** of the epithelial cells into the mesenchyme the midline epithelial seam is disrupted.

2. As palatal growth the seam first **thins down** and then breaks up into discrete islands of epithelial cells.

3. The basal lamina surrounding these cells then is lost, the **epithelial cells lose their epithelial characteristics** and assume mesenchyme like characteristics hence the mesenchymal cores are continuous with each other to allow proper fusion.

4. There is **apoptosis** of epithelial cells.

Later, ossification begins in the anterior part (forming the hard palate) and remains membranous posteriorly (forming the soft palate).



Palatal shelf elevation

Extrinsic forces:

- 1 descent and forward movement of the tongue
- 2 growth of the mandible.

Intrinsic forces:

- 1 hydration of extracellular matrix component as a result of mesenchymal cell activities.
Hyaluronan is a GAG involved in shelf elevation- increased tissue turgor pressure.
- 2 mesenchymal cell activity

Clinical Consideration- Cleft Palate



Disturbances during palatogenesis

- 1 delayed shelf elevation
- 2 failure of elevation
- 3 failure of shelf fusion
- 4 lack of degeneration of the midline epithelial seam

Multifactorial origin/teratogens

- 1 Environmental (malnutrition, folic acid def, radiation exposure, smoking, ingestion of alcohol or isotretinoin)
- 2 Genetic
- 3 Infectious (rubella)

Syndromic

- 1 Pierre Robin's sequence
- 2 Treacher Collins' syndrome

Table 18.1 Clefts of the upper lip and of the palate

Cleft lip	Cleft palate
Unilateral or bilateral	Classification with reference to incisive fossa – anterior anomalies (primary palate) and posterior anomalies (secondary palate)
With or without cleft palate	
1 : 1000 births	1 : 2500 births
Mostly male infants	More common in females
67% with multifactorial aetiology	41% with multifactorial aetiology
Result of failure of fusion of mesenchymes in facial processes	Result of failure of elevation of palatal shelves (major clefts) or failure of fusion of shelves (minor clefts)